Distributed intelligence and data in public and private networks

R P Swale and D R Chesterman

This paper considers the relationship between intelligence and applications embodied within both public and private telecommunications networks. It identifies the emergence of intelligent network (IN) and computer/telephony integration (CTI) as viable and synergistic control architectures for each environment where the basic communications control mechanisms are becoming increasingly centred on computing platforms external to the basic switching fabric. It also considers aspects of why these systems are migrating towards reliance upon a distributed processing approach and suggests that this is a logical progression for current design evolution.

1. Introduction

In the early days of telecommunications networks — and telephone networks in particular — call completion was dependent upon manual switching and there was a strong reliance upon human intervention to deliver the end service. Whilst this structure afforded the opportunity of a near infinite range of supplementary services, it was also open to less attractive features such as inconsistency in service and feature delivery — not to mention fraudulent misuse. The adoption of electro-mechanical switching resolved many of these issues. However, while this produced an inherently more consistent network and associated service offering, it did so at the expense of the 'intelligence' employed within the network.

The application of digital computing technology to telecommunications network systems has afforded the opportunity of developing a wider range of services and facilities. These developments may be considered to be in some way replacing the previously lost 'network intelligence' and are reintroducing scope for network operators to offer an increasing range of enhanced services. Indeed, in recent times public network operators have sought to support their customers' needs for using telecommunications as a way of obtaining the 'edge' in an increasingly competitive market place. This has largely been achieved through network based services, such as Freefone. A key feature of the realisation of these services is their strong reliance upon service related data and the manipulation of that data in the operation of the service by 'intelligence' functions. This is the essence of present-day network intelligence. While it is possible that for many services all the necessary information could be held and processed by a single modern digital exchange, it is apparent that there are significant operational difficulties with this approach for anything other than the most trivial of network topologies. For example (ignoring the concept of 'overlay networks'), in a reasonably sized network employing a number of switches, each switch in the network would need to retain both the number translation and charging data for each Freefone number operated. Furthermore, any updating of the service data is likely to involve data changes to all network switches which will invariably take time to complete, especially if a number of different switch types are used in the network.

The effects of increased reliance upon service data, the need for manipulation of that data by the network and the distribution of the data throughout the network topology has been the motivation behind the 'intelligent network' (IN) techniques discussed elsewhere in this journal [1, 2]. However, solutions based upon these principles to date have tended to be implemented on a platform per service basis which has largely migrated the issues of service logic consistency and minimisation of data duplication away from the switching domain and into the domain of service control platforms realised by computing resources. Furthermore whilst the IN approach has emerged as a desirable architecture for the public network in the delivery of advanced services, private telecommunications systems have seen the complementary emergence of computer/telephony integration (CTI) as the route to providing tighter coupling between business information systems and 'front-office' call handling in the quest for greater business efficiency.

This paper provides an overview of both the public and private domains and presents a view as to the common issues each faces in terms of the computing problems which must be solved. This paper therefore presents an assessment of aspects of the computing issues for realising both public and private intelligent networks. As public IN systems have been discussed elsewhere this paper explores private systems and CTI in greater depth.

2. Computers and telecommunications

The network intelligence playing field may be considered to essentially consist of computer based intelligences linked together by communications capabilities. Whilst these two fields of computation and telecommunications have originated separately, it is apparent that their relationship has grown somewhat unclear over the past 10—30 years. For instance, the introduction of common control (computer-based) switching systems within the telephone network and the development of complex communications systems (such as Ethernet) within the computing environment may be seen as examples of this theory in practice. It has also been observed that developments within the computing field have in many ways preceded those in the public telecommunications field — as confirmed by a consideration of computer system architectures [3]. From such material it is apparent that there is a trend towards processing capabilities migrating to the edges of the physical architectures employed. This may be envisioned using a subjective tool termed an 'intelligence distribution graph' as shown in Fig 1 which presents characteristics for centralised host systems, network interconnected host systems and also the distributed computing systems typified by client/server architectures.

However, when making such analogies, it is important to recognise that the stringent, non-functional characteristics of telecommunications systems do not tend to be matched by commercial computing platforms. As a consequence, while much may be learned by studying computer system architectures, the solutions proposed may not directly translate into the telecommunications environment. Even so, it is apparent that, as computing platforms evolve, there will be an increased reliance upon and integration with telecommunications facilities to meet the demand for both increased access to data and other resources and the sharing of computational load. The migration towards distributed systems architectures will therefore force alignment between the non-functional characteristics of these two domains.

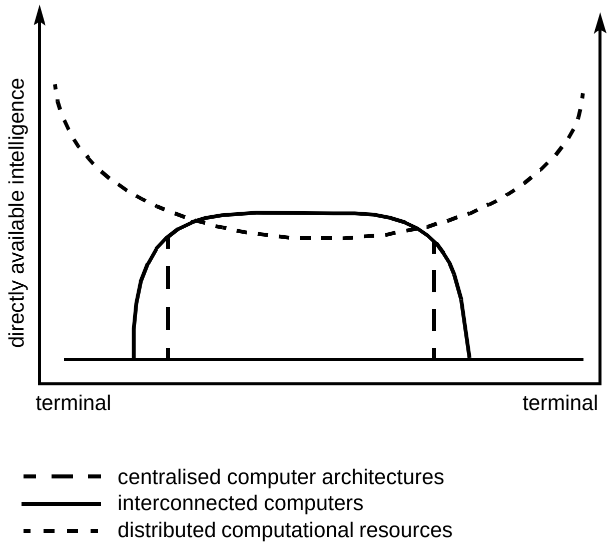

Fig 1 Intelligence distribution graphs for various system architectures.

3. Distribution of intelligence and data in public networks

Within modern public telephone networks, network services are currently delivered either through ser-vice logic embedded within switching systems or through the use of the emerging IN-type control architectures. The following discussion presents a consideration of these architectures from the point of view of how service logic and data is distributed from a control architecture view point.

3.1 Non-IN approach

Until the development of IN-type control architectures, network-based intelligent call-handling services were generally implemented as part of the software within the stored program control (SPC) telephone exchanges used to realise the public network. Within the multivendor switching environments operated by most telcos throughout the world, this causes a number of significant problems which are well understood and have been a motivating factor for adopting IN control architectures. These problems are mainly centred on the use of multiple switch types which give rise to variations in service logic implementations and also the reliance upon switch vendors to deliver the service logic as part of exchange software drops. These effects combine to render a consistent overall service offering both difficult to achieve and time consuming to deploy. A further disadvantage of this approach, which results from embedding the service logic within the switching system, is that the service data is also bound to the switching system. In the case where several switches are used this may cause service data to be replicated across several switches. This causes management overheads in administering the service, consumes more exchange resources, such as memory, due to data duplication and creates difficulties in implementing subsequent services. This last point is important when personal mobility services are considered.

3.2 IN structured networks

Most major telcos throughout the world have experimented with IN-type control architectures in trying to overcome some of the deficiencies found with switch-based service logic and data. Indeed, IN techniques have proved their worth in providing consistent service offerings through the provision of services independent of the underlying switching systems. However, since few public networks fully support IN control systems from all exchanges, difficulties remain in that customer service logic and data is now distributed across both SPC and IN control platforms. In this sense IN is still far from being the panacea it could be. This is compounded by the fact that at present IN platforms tend to be established for particular services as they are identified and brought to market, which leads to scope for further duplication of customer service data across several platforms.

3.3 Observations

Within the public network, there is a major drive towards migrating service logic away from the core network switching systems in order to provide a more flexible approach to service provision. To avoid a simple migration of the problem from the switching fabric to the IN control platforms, there is the need to maintain a single consistent view of the service logic from two aspects. Firstly, there is the need to ensure that all parts of the switching fabric can access the same service logic and data to support possible services such as personal numbering and personal mobility. Secondly, there is the need to ensure that associated operational support systems (OSS) can obtain a single logical view of operating services to provide a consistent view of services to the various ‘front-office’ systems. With the potentially enormous volumes of data implied by a full public network, both of these factors imply the adoption of a distributed systems approach to the platform solution adopted.

4. Distribution of intelligence and data in private networks

As with public telecommunications networks, private networks have emerged from a background of manual switching. The drive towards providing enhanced facilities has seen migration towards the use of computer-based systems to deliver the required service logic. However, it is considered that the drivers within the private arena are somewhat different to the public network in that the profusion of facilities has largely exceeded saturation point and the development path now focuses upon applications rather than specific facilities — as will become apparent from the following discussion.

4.1 Embedded switch functions

Within private networks and switching systems, there is a parallel with the non-IN based solutions of public networks where facilities are embedded in the switch platforms. This has led to the proliferation of a wide range of facilities on private switching systems and the use of enhanced private network signalling systems, such as digital private network signalling systems (DPNSS), to allow many facilities to operate throughout a private network. Unfortunately, it is also now becoming widely recognised that many of the enhanced facilities are seldom, if at all, used — largely due to the user-unfriendly interface supported, which is based upon raw dialled digits and the incompatibilities of features and feature invocation from one manufacturer to another. Just as with the public environment, where there has been a drive towards removing service logic from switching systems to allow network services to be more rapidly created and delivered to the market-place, customer premises equipment (CPE) has largely been driven by the ability to provide ever more powerful facilities as a means of product differentiation.

4.2 Computer/telephony integration

Since the advent of digital telephony more than a decade ago, the convergence of computing and telecommunications in CPE has been an obvious and much heralded concept. However, discussion which is centred on the physical convergence of the LAN and the PBX has generated little interest amongst potential users, since they are generally not concerned about the physical infrastructure on which their services are delivered. Instead, it is the convergence of applications, and the appearance of standards to support this, which is fuelling the current rapid growth in computer/telephony integration (CTI).

There are two basic approaches to CTI, referred to as 1st and 3rd party CTI. These are discussed here.

1st party CTI

1st party CTI is characterised by the fact that integration takes place at an individual user’s telephony ports, the computer (typically a PC) being given access to the signalling capabilities of that port. Figure 2 illustrates a number of possible configurations:

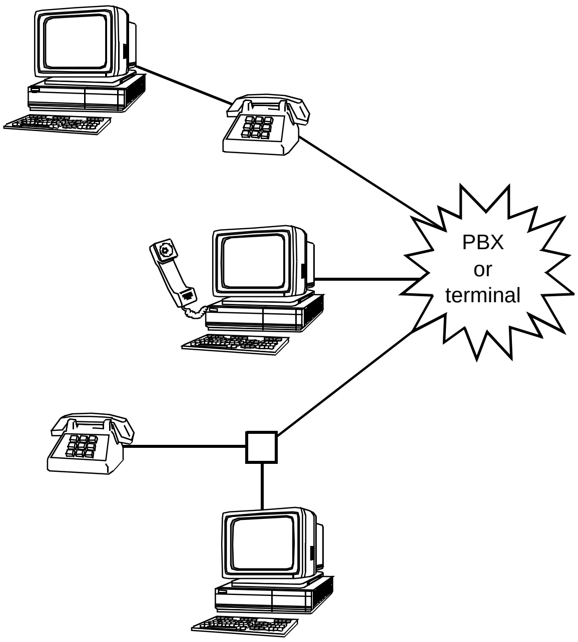

Fig 2 1st party CTI.

- the first of these is typical of a PBX digital featurephone, in which the phone provides a serial data port which gives access to both the signalling capabilities of the port, and the user (voice) channel for data transfer — while the PC may be used to make, answer and log calls, the handset of the featurephone would normally be used for speech,

- in the second configuration, the PC subsumes the full functionality of the phone,

- the final configuration is typically used on analogue extension ports, and uses a 'smart box' to allow the telephony signalling to be presented to the PC's serial data port — the 'smart box' may also include modem functionality for data transmission with speech provided via the phone's handset, but both the phone and the PC may take or share control of a call.

It can be seen that 1st party CTI either provides the PC with a serial link for telephony control, or requires the addition of a telephony card to the PC. There are a large number of such cards now available, providing analogue (2-wire or 4-wire), ISDN basic rate (144 kbit/s), primary rate (G.703, 2.048 Mbit/s), or asynchronous transfer mode (ATM) interfaces. Some cards have multiple ports and supported signalling systems include L/D, MF, CLASS, ISDN (Q.9xx), ISDN30 (DASS2), DPNSS, and even CCITT No 7. It is also possible to include multiple cards within a PC chassis, and proprietary or *de facto* standards have emerged for the interconnection of speech channels between cards within the PC, typically on 2.048 MBit/s ribbon connectors. Typical of these are SCbus (part of Dialogic's signal computing system architecture — SCSA), and MVIP (multivendor integration protocol). Additional functionality available on PC cards includes modem facilities, fax encoding, voice and video codecs, speech compression, and 64 kbit/s switching.

Applications for 1st party CTI are almost unlimited, and currently include such diverse facilities as screen-based featurephone, auto-dial from a screen directory, video-phone, PC-based answerphone, integrated e-mail and voice-mail, facsimile services, and call-logging. Full multimedia communications via ISDN is available, and is exemplified by BT's VC8000 product.

The availability of new applications is likely to be considerably accelerated by the inclusion of a telephony applications programming interface (TAPI) in Microsoft's Windows 95 (Chicago) [4]. This enhances the normal Windows API with a set of functions which may be called to enable telephony events to be monitored, and telephony commands to be given. Figure 3 illustrates the TAPI environment, showing a variety of applications accessing the Windows libraries, with third party 'service providers' giving access to the capabilities of various PC cards and network services.

Figure 4 illustrates how voice-mail might be integrated with e-mail in a typical application (in this case, Microsoft's 'Mail'). The voice-mail is distinguished by the phone icon.

There are a number of points to note about 1st Party CTI.

- Applications are not limited to the desk top. The availability of primary rate cards makes the provision of centralised or networked facilities a practical reality. Indeed, BT's own audio and video conference services are provided in this way. The distinction between 1st and 3rd Party CTI (see below) becomes very blurred in such applications.

- 1st party CTI relies on normal telephony or ISDN ports, and can therefore be applied to any system (e.g. PBX, Centrex, PSTN, etc) without enhancement to the switch.

- A LAN is optimised for bursty, asynchronous, high-bandwidth communications, and therefore provides the most optimal solution for the transport of applications data. Equally, the telephone system has been optimised for delay-sensitive, and constant low-bandwidth communication. Thus, in a normal office environment it is normal to provide both media (as illustrated in Fig 2), with the telephone/ISDN networks only being used for data communications where the user is working remotely (e.g. teleworking). However, technologies such as ATM and isochronous ethernet (isoENET) are becoming available in the CPE environment, and these will make integration at the transport level more common in the future.

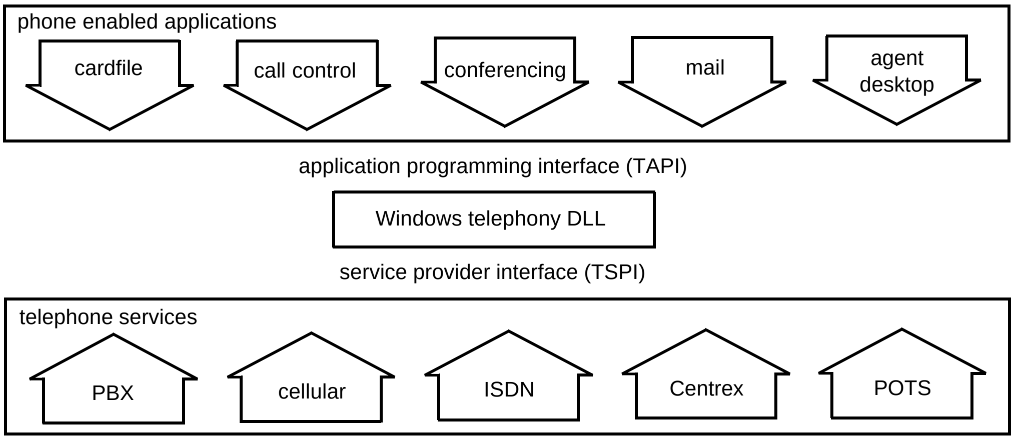

Fig 3 The TAPI architecture.

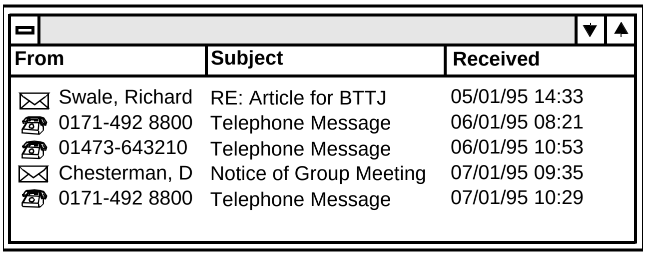

Fig 4 Integrated voice and e-mail.

- The wide variety of interface cards available make it possible to build a complete PBX in a PC, and at least one such system is now being marketed. While the benefits of such an approach for simple telephony are doubtful, the advent of true multimedia PCs combined with communications media such as ATM would seem to point to a future where the LAN hub and PBX become one.

3rd Party CTI

Figure 5 illustrates a basic 3rd party CTI architecture, and it can be seen that this very closely parallels an intelligent network. For this reason, 3rd Party CTI is sometimes referred to as ‘private IN’.

These systems rely on the addition to the telephony switch (PBX, ACD, or Centrex) of a CTI port, which provides a telephony command and status link to a host computer. The capabilities provided via this link are similar to those available through 1st Party CTI, these being the ability to monitor activities on the switch ports, and to assume the control of calls through primitives such as ‘Make_Call’, Clear_Call’, ‘Transfer_Call’, etc. However, there are two important additions:

Fig 5 3rd party CTI.

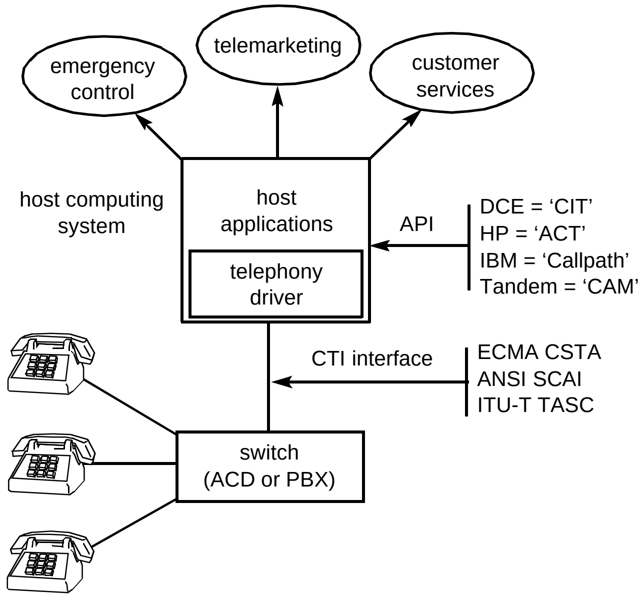

Fig 5 3rd party CTI.

- it is often possible to take control of the routeing of a call before it has been delivered to a port,

- the host computer system has the opportunity to monitor and control **all** calls and devices on the switch.

3rd Party CTI messages are carried on a variety of systems, with X.25, Ethernet and (to a lesser extent) ISDN layer 2 being the most common. Most PBX manufacturers have defined their own proprietary message sets, although several standards have now been ratified. These are:

- CSTA — defined by task group 11 of technical committee 32 of ECMS — although this is a European standard, it has won the active support of the majority of major US switch and computer vendors,
- SCAI — defined by ANSI (T1.626), this standard tends to be favoured by those providing virtual private services (e.g. the Regional Bell Operating Companies (RBOCs)),

- TASC from the ITU — this work was started in an attempt to bring the work of ECMA and ANSI together into a single standard; although the first issue of TASC has been ratified, work on this standard has now, however, been suspended.

Tables 1 and 2 compare the capabilities of the CSTA and SCAI protocols, and also includes the equivalent INAP primitives. It should be noted that these standards are evolving, so the information should not be taken as a definitive statement of current capabilities. From Tables 1 and 2 it can be seen that SCAI is currently less feature-rich than CSTA. However, unlike CSTA, SCAI does mandate a call model, which makes its behaviour more predictable as applications are moved between different vendors' switches. SCAI was designed with an emphasis on the provision of network services, and is considered by some to be the more secure protocol. However, a full implementation of the SCAI standard has never been brought to market, although a number of partial implementation do exist.

Most CTI applications are developed on a telephony API supplied by the computer vendor. These are invariably proprietary, since no standards currently exist in this area.

Such APIs have the effect of shielding the applications developer from the detail of the physical CTI interface, and, in doing so, make the application portable across the range of switches supported by the chosen computer vendor. Unfortunately, this switch independence is gained at the expense of lock-in to the computer manufacturer, and, to an extent, this negates the advantages of standardisation of the CTI interface.

Table 1 Showing a mapping, at the functional level, of the various host to switch telephony services defined within SCAI, CSTA and CS-1. (Note that services are likely to be similar, rather than equivalent.)

SACI services [host→ switch]	**CSTA services** [host→ switch]	**CS-1 operations** [SCF→ SSF, SDF and SRF]
Initiate_Monitor	Monitor_Start_Service	Request report BCSM event
(parameter defines event 'mask')	(parameter defines event 'mask')	Request every status change report
Change_Monitor_Filter	Change_Monitor_Filter	Request 1st status match report
		Request notification charging event
Query_Monitor		
	Query_Device	Request current status report
Cancel_Monitor	Monitor_Stop_Service	Cancel status report request
		Cancel
		Activate service filtering
Make_Call	Make_Call	Initiate call attempt
		Connect
		Establish temporary connection
Route_Selected		Select route
		Select facility
		Connect to resource
		Collect information
		Analyse information
		Apply charging
Answer_Call	Answer_Call	
Clear_Call	Clear_Call	Release call
		Disconnect forward connection
		Continue
		Call information request
		Furnish charging information
		Send charging information
Predictive_Make_Call	Make_Predictive_Call	
Conference_Existing_Calls	Conference_Call	
Drop_Conference_Party	Clear_Connection	
Single_Step_Transfer	Consultation_Call	
	followed by Transfer_Call	
Consultation_Transfer	Transfer_Call	
Hold_Call	Hold_Call	Hold call in network (similar)
Retrieve_Call	Retrieve_Call	
Query_Feature		
Set_Feature	Set_Feature	
Routeing_Toggle		
	System_Status	
	Event_Report	
	Call_Completion_Service	
	Reconnect_Call	
	Divert_Call	
	Snapshot_Call	
	Snapshot_Device	
	Alternate_Call	
	Escape_Service	
		Call gap
		Reset timer
	System_Status	Activity test
		Play announcement (SRF)
		Prompt and collect user info (SRF)
		Query (SDF)
		Update data (SDF)

Tabel 2 Showing a mapping, at the functional level, of the various switch to host telephony services, events and triggers defined within SCAI, CSTA and CS-1. (Note that services and events are likely to be similar, rather than equivalent. In particular, the call control function will suspend processing when it reports an CS-1 trigger, whereas a CTI call control will continue after reporting an event.)

SCAI services and events [switch→ host]	**CSTA services and events** [switch→ host]	**CS-1 services and triggers** [SSF, SDF and SRF→ SCF]
Monitor_Report information		
Roupte_Request	Route_Request	Analysed information
	Route_Used	
	Route_Select	
	Route_End	
	Re_Route	
		Specialised resource report(SRF)
		SDF response (SDF)
		Event report BCSM
		(parameter indicates event)
		Initial DP
Service_Initiated	Service_Initiated	
		Originating attempt authorised
Call_Originated	Originated	
		Collected information
Call_Delivered	Delivered	
Call_Arrived		
Call_Received	Delivered	O Answer, T Answer*
Call_Established	Established	
		O Midcall, T Midcall*
Call_Cleared	Call_Cleared	O Disconnect, T Disconnect*
		Event notification charging
Call_Failed	Failed	O No Answer, T No Answer*
(parameter indicates cause)	(parameter indicates cause)	O called party busy, T called party busy*
		Route select failure
		Call information report
Agent_Logged_On	Logged_On	
Agent_Logged_Off	Logged_Off	
Agent_Ready	Ready	
Agent_Not_Ready	Not_Ready	
Agent_Working_Not_Ready	Work_Not_Ready	
Agent_Working_Ready	Work_Ready	
		Service filter response
Calls_Conferenced	Conferenced	
Conference_Party_Dropped	Connection_Cleared	
Call_Transferred	Transferred	
Call_Held	Held	
Call_Retrieved	Retrieved	
Call_Diverted	Diverted	
Network_Reached	Network_Reached	
		Status report
	Queued	
	Back_In_Service	
	Out_Of_Service	
	Call_Information	
	Do_Not_Disturb	
	Forwarding	
	Message_Waiting	
		* (The O and T prefixes indicate whether the event is reported at the originating or terminating end of a call)

Some software suppliers have adopted the approach of developing an API which exactly mirrors the CSTA standard. Notable amongst these is Novell, whose product (TSAPI) also follows the trend in 3rd Party CTI systems towards client/server architectures [4].

Figure 6 illustrates a typical client/server CTI installation, and it is seen that a telephony server (possibly duplicated for increased availability) distributes the switch's capabilities to a number of distributed client applications. These might be providing system-wide facilities (such as call-logging, call distribution, or routeing services), or could be running call-handling applications on a user's workstation. In the latter case, these applications will look very much like those which would be provided through 1st Party CTI. However, they will benefit from reduced costs in a large installation, since the cost of the server is likely to be outweighed by the saving of a telephony card per PC.

Currently, most 3rd Party CTI systems are used in call centre applications, usually for inbound or outbound tele-marketing operations [5]. An examination of the commands and events supported by CSTA and SCAI (such as Agent Logged-In/Logged-Out/Ready/Not-Ready) clearly indicates that the standards have been heavily influenced by this very important and fast-frowing market. The benefits brought to a call centre through the inclusion of CTI include:

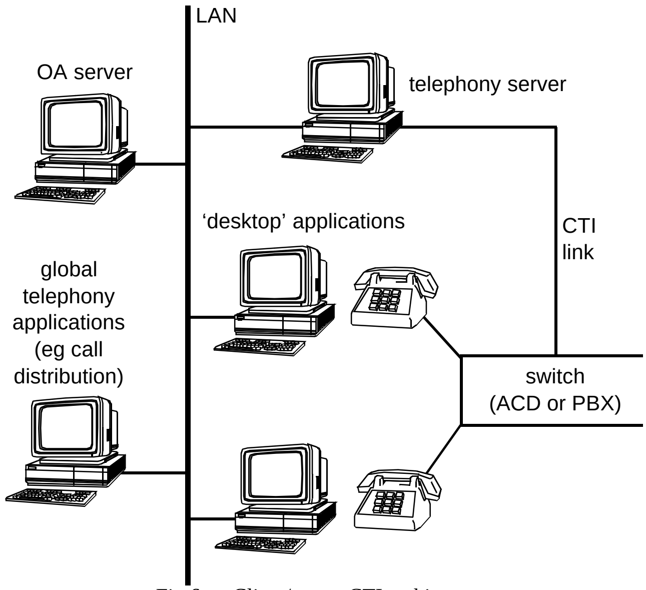

Fig 6 Client/server CTI architecture.

- the ability to use multi-skilled agents — by using CTI to present the dialled number to the telemarketing/ scripting application, the agent will be presented with the appropriate screens and scripts for handling the requested service (e.g. sales, enquiries, service),

- the presentation of Calling Line Identity (CLI) to the application allows the automatic selection of t he (likely) customer's records,

- the compilation of sophisticated call-handling statistics (the life-blood of the call centre manager),

- the ability to be able to simultaneously transfer a call and the associated data (customer records, order forms, etc).

These in turn lead to cost savings through increased efficiency, improved customer/caller satisfaction, and increased job satisfaction for the agents.

Although call centres have been the primary focus for the development of 3rd Party CTI, the capabilities of these protocols are generic enough to be able to build a wide variety of services, and there is a growing interest in services such as 'hot desking' and 'personal numbering'.

4.3 Observations

With the massive variety of facilities available on modern private switching systems largely exceeding an individual user's ability to comprehend, let alone use them, there has been a continued drive up the value chain into enabling a business's 'back-end' data to be tightly coupled into 'front-end' office applications. In many ways this can be seen to be indicative of where IN-based public networks will end up once the major services have been defined and deployed. However, this may be a somewhat restricted view of the future since there is a largely untapped potential in bringing the two technologies of IN and CTI together [6, 7] and exploiting the relative strengths of each, particularly as each environment makes use of increasingly similar distributed computing platforms.

5. Relationships between intelligent public and private networks

As already indicated, CTI protocols are generic enough to be able to build a private equivalent of many IN call-handling services. The removal of restrictions on network to network calls also mean that such services are not necessarily restricted to the private network domain, and there are examples of 3rd party service providers using CPE to deliver facilities which would normally be considered to be in the domain of the telco. Clearly, these services will suffer an economic penalty when they require calls to be tromboned out of and back into the network, but this is not always the case. As an example, consider a 'personal numbering' service implemented on a corporate network. Given that the personal number would ideally apply to all calls, including those which originated within the private network, then an analysis of the possible call sources and destinations will indicate that for most users there will be little difference in the call charge costs for the private and public implementations. However, the private implementation has the potential benefit of being easier to integrate with other office automation systems, such as diaries and schedulers.

A comparison of the CTI and IN capabilities in Tables 1 and 2 indicate the similarities in the functionality offered in both environments. However, CTI in its various proprietary forms is a more mature technology, particularly in the USA where it is the fundamental building block of many very large call-handling operations. Also it will not be easy for IN to close the gap on CTI, since the CPE environment is one in which responsiveness and innovation are easy to achieve. This environment is largely unregulated, is not constrained by licence restrictions, is fiercely competitive, does not incur the availability, process, billing and security overheads which apply to a network operator, uses cheap technology, and is a very easy area within which to develop.

However, the CPE and network environments should not be regarded as competing with each other, and a closer examination of the CTI and IN protocols presented in Tables 1 and 2 shows that they have been specified with entirely different objectives. The role of the network is to deliver calls, and the role of CPE is to handle them efficiently and in a user-friendly way at the network termination point. Thus a protocol like INAP does not assume capabilities such as transfer, call distribution, predictive dialling, etc, within the switching fabric, since these are in themselves IN services. On the other hand, whilst CTI provides a pragmatic interface to a switch or network which is already highly featured, it lacks the range of security, charging and other features required for control of a public network. By contrast, the IN has a range of functionality essential to the service provider (operations and maintenance, mobility, charging, etc) which is either poorly specified, or non-existent in CTI. CTI and IN technologies may therefore be considered to present orthogonal control views on to the public network with IN presenting a 'vertical' view for call delivery services and CTI presenting a 'horizontal' view for post-call delivery call-handling applications.

If either the network or CPE tries to compete in the domain of the other, then it is likely to lose. Abbreviated dialling is a simple but good example of a service which is far better provided from CPE than from the network. Equally, though, incoming call distribution between several locations is better performed from within the public network as this avoids CPE 'tromboning' calls back into the public network, and allows truly global distribution.

To an extent, the threats to the network from intelligent CPE are balanced by such opportunities. To take the earlier example, a corporate network provided via Centrex services will be able to provide a truly universal personal numbering capability to its users without ever incurring double-switching charges. It is also seen that call distribution is another significant opportunity. Many large telemarketing operations are distributed across a number of sites, and there is an increasing interest in the use of teleworking in this environment. The ubiquity of the network provides the opportunity for truly global call distribution, with the efficiency and management benefits of being able to control this through a single (logical) queue.

The greatest benefits are achieved when the true roles of network and CPE are respected, and each is used as a component in the delivery of the total solution to the customer. However, to achieve these benefits, there is a need to look beyond the purely telephony signalling capabilities of the access network.

5.1 Relationships between public and private IN

It is apparent from a consideration of the architectures of both public and private IN systems that there are considerable similarities between the major elements of each architecture. At a simplistic level, each architecture comprises a switching system which is controlled by an external computing platform. To this may be added a number of supporting resources such as communications peripherals (voice response units, facsimile modems, etc) and database systems. These features are shown in Fig 7.

As there are such striking similarities between these two architectures, there are a number of potential opportunities

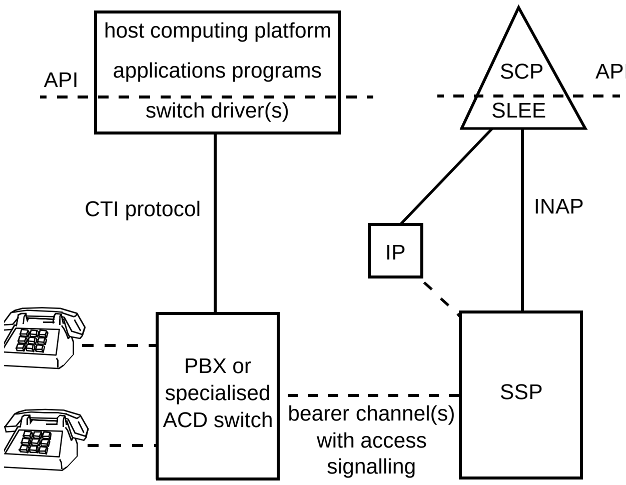

Fig 7 Comparison of public and private IN architectures.

for providing interconnection. The usefulness or otherwise of these interconnection opportunities is best illustrated through some examples.

Simple interconnection

In this case, which is typical of current call centre systems, only bearer-level relationships exist between the public and private IN platforms. Calls are processed in the public network before being delivered to the network termination point. Where additional call-related information is made available through the bearer signalling system, e.g. calling line identity, this information may be subsequently used by private IN systems to perform further independent processing on the call after it is delivered to the network termination point, but before the call is answered.

SCP access to private database

When executing service logic, a public network SCP platform will typically interact with a variety of data which is usually held in an on-line accessible database. A typical call flow through the platform will require one or more accesses to this data to route the call; for example a simple number translation application may 'dip' the database to obtain the translation of the dialled number. It is possible to extend this interaction beyond the public network platform's database to an interaction between the SCP and a database held within a private network. In this case, it is apparent that the access time to the database is a critical factor in the overall performance of the platform. Other factors which will be complicated by extending the interface between the database and the SCP to a database in the private domain include:

- security,
- reliability,
- availability.

These factors combine to make this particular interface potentially difficult to realise except where the service being realised can tolerate the delays and where the SCP technology can support the numbers of customers required. This last point is important in this context because a ‘thread’ of control may have to be maintained for a longer time period within the SCP for an external database access to the private database than would occur for a ‘local’ database access which implies the need for greater computational resources.

Host control of public switch

An alternative approach to providing private domain control over call delivery from the public network is to allow a private network host to control the calls delivered at a given network termination point. At the private network end, the structure becomes identical to a 3rd party CTI structure with the PBX replaced by the local public switch and so this would be relatively simple to implement. Conversely at the public network end there is the need to ensure that the control partition is adequately policed to avoid the control interface acting upon calls which it has no right to manipulate (it is interesting to compare this with the early days of telephony where suspected partiality resulted in Strowger switching systems). The additional control flows between the local switch and the private host platform are also likely to generate a significant additional computational load for the switch processor (current services such as ISDN D-channel packet data and IN control signalling are notoriously computationally intensive for exchange processors) — particularly where several customers on one exchange require this type of service.

5.2 Similarities between CTI and IN architectures

The ECMA CSTA specification draws upon the open distributed processing work being conducted by ISO in identifying relevant functional domains [8]. With the view that the switch is the controlled entity and the computing platform is the controlling entity which hosts the various controlling applications, it is apparent that the computing platform must take responsibility for resolving the control flows to the various applications. For example, consider the case where a system uses two Freefone numbers — one for general customer services and sales and the other for servicing and maintenance. Incoming calls will cause ‘route requests’ to originate from the switch in response to received calls and it will be the responsibility of the computing platform to resolve which ‘service logic’ instance should be executed for each call. In this example the lower layers of the computing platform could route to either the service logic for customer services or the service logic for servicing and maintenance dependent upon the called line identity.

In the case of public IN platforms, elements of the software functions which provide this routeing function are embodied in the service switching and service control functional entities identified in ITU recommendation Q.1214 [9]. By contrast CTI systems tend not to identify this functionality in such a generic manner as it is generally encapsulated within the proprietary applications environments which vendors provide on top of the basic CTI interfaces as indicated in Fig 5.

However, there is a high degree of similarity between the control platforms for IN and CTI systems in that they must provide the distribution of control flows between the interface to the switch and the call-handling application. This call-handling application could also be on a separate processor to that supporting the switch interface and in a CTI environment this is almost certain to be the case. This could present difficulties for applications developers as they have to know the specific structure of the control platform in order to develop the call-time applications. This is clearly undesirable and fortunately this is a sufficiently generic distributed-systems problem to ensure that solutions either are available or will become available.

6. Conclusions

Although public and private telecommunications networks have common roots in manual switching, they have evolved into IN-type architectures for very different reasons. In the case of public networks this has been seen to be due to the need to provide a more flexible route to offering an increasing range of call delivery and handling services. Public IN developments have accordingly been directed at satisfying the need to support call manipulation at the pre-call delivery stage, with a relatively find degree of granularity. Conversely private systems have been driven by the user requirements of greater integration with both new and existing office/business automation and business information applications. CTI developments have therefore been driven by a slightly different set of requirements and in some cases the associated control protocols have been produced at a more coarse degree of granularity than those employed for public IN. There are also some special concepts embodied within CTI protocols which have no true parallel within public IN. These include the concept of supporting ‘agents;’ in the context of telemarketing systems. Applications programming interfaces within CTI systems are also progressing towards de factor standards such as Microsoft’s TAPI, whereas the near equivalent call-time SLEE interfaces on public SCP platforms remain largely proprietary and closed. In contrast, public IN systems have a greater integral support for operations and management than that offered by CTI system.

Set against this background of contrasting drivers for public and private IN systems, there are striking architectural similarities between the two. These have been seen to establish similar requirements for the controlling computing platforms [8, 9] in terms of the need to support multiple call time applications to the switch side while maintaining a single consistent view for application developers and system operators. Both of these effects have combined to drive the computing platform solutions towards the use of a distributed systems approach to avoid data duplication and application incompatibilities. As both public and private systems migrate to such distributed control system architectures, there will clearly be increased scope for interworking and at some point in the future it may be practicable for the public and private systems to appear to more fully participate as a single holistic platform.

Acknowledgements

The authors would like to acknowledge the contribution to this article from Gary Crook and their other colleagues in the network intelligence engineering centre.

Windows and TAPI are trademarks of the Microsoft Corporation, and TSAPI is a trademark of Novell Inc. All other trademarks acknowledged.

Appendix

List of acronyms

ACD	automatic call distribution
API	applications programme interface
ANSI	American National Standards Institute
ATM	asynchronous transfer mode
BCSM	basic call state machine
CLI	calling line identity
CPE	customer premises equipment
CSTA	Computer Supported Telephony Applications — the ECMA standard for CTI
CTI	computer/telephony integration — this is the currently accepted acronym, though the concept has been referred to by a number of other terms, e.g. computer supported telephony (CST), computer integrated telephony (CIT), host control integration (HCI)
DLL	dynamic link library
ECMA	European Computer Manufacturers Association
ETSI	European Telecommunications Standards Institute
INAP	intelligent networks application protocol
ITU	International Telecommunications Union
IP	intelligent peripheral
LAN	local area network
MVIP	multivendor integration protocol — a standard for multi-64 kbit/s channel interconnection between PC expansion boards, independent of the PC bus
P(A)BX	private (automatic) branch exchange
POTS	plain old telephony service
RBOC	Regional Bell Operating Company
SCF	service control function
SCP	service control point
SDF	service data function
SDP	service data point
SRF	specialised resource function
SSF	service switching function
SSP	service switching point
SCAI	Switch-Computer Applications Interface — the ANSI standard for CTI
SCSA	Signal Computing System Architecture — an architecture for the integration of voice peripherals in a PC, which includes an API and the specification of a multi-64 kbit/s inter-connection between expansion boards (although specified by Dialogic, a number of other vendors are now following this 'standard')
TAPI	Microsoft telephony applications programming interface
TASC	Telecommunication Applications for Switches and Computers — the ITU standard for CTI.
TSPI	telephony service provider interface
TSAPI	Novell telephony services applications programming interface

References

1 Abernethy T W and Munday C A: 'Intelligent networks, standards and services', BT Technol J, 13, No 2, pp 9—20 (April 1995).

2 Bale M C: Signalling in the intelligent network', BT Technol J, 13, No 2, pp 30—42 (April 1995).

3 Tanenbaum A S: 'Structured computer organisation', 2nd Edition, Prentice-Hall International (1992).

4 Wilson N: 'PC players set the CTI standard', Telecommunications, International Edition (October 1994).

5 Bonner M: 'Call centres — doing better business by telephone', Journal of the Institution of British Telecommunications Engineers, 13, Part 2 (July 1994).

6 Swale R P: 'Virtual networks of the future — converging public and private IN', IEE Colloquium on Virtual Networking (October 1993).

7 Swale R P et al: 'Convergence of public and private IN', Proceedings of the 2nd International Conference on Intelligence in Networks, Bordeaux, France (March 1992).

8 Computer supported telecommunications applications — ECMA 179.

9 ITU-T recommendation Q.1214: 'Distributed functional plane for intelligent CS-1'.

Richard Swale began his career in 1981 as an apprentice at BT Laboratories. He read Electronic Engineering at the University of Kent and graduated in 1988 with a first class honours degree. During the following three years he worked on the design of specialised computer systems and customer data networks within BT's operating divisions.

In 1991 he rejoined BT Laboratories to study the interaction between the IN and complex CPE, and now leads a group studying the application of distributed computing to network intelligence platforms.

He is a Chartered Engineer and a member of the IEE.

Doug Chesterman joined BT in 1973, after graduating from the University of Kent at Canterbury with an honours degree in electronics. After a 5-year period in which he contributed to the network planning department's early strategies for the introduction of digital switching systems, he moved to BT Laboratories and joined the System X development project. Here he took a leading role in the design and implementation of a new call processing architecture, which was subsequently adopted by the System X prime contractor.

He currently leads a group which specialises in intelligent CPE and access systems, and in understanding the relationships between these and core network intelligence.